Architectural Fidelity

Empirical Observations on Data Locality & Memory Latency

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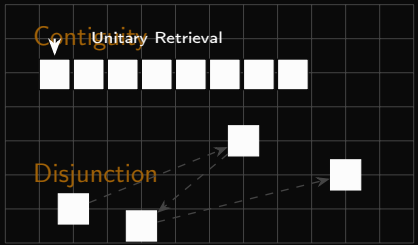
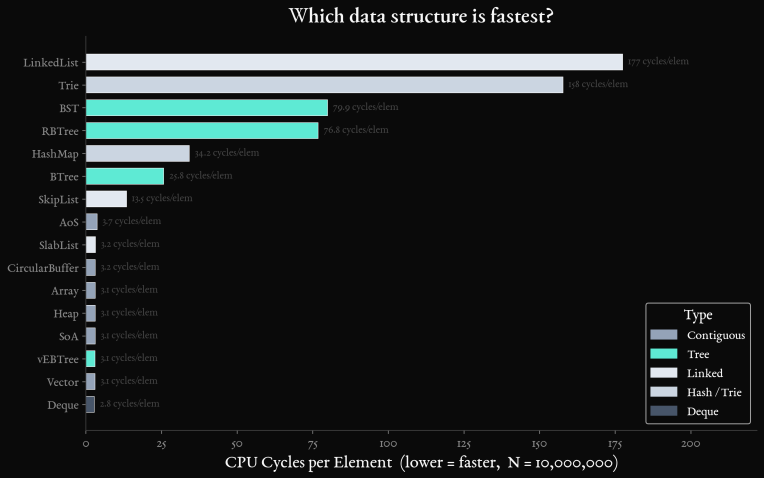
The Spatial Premise



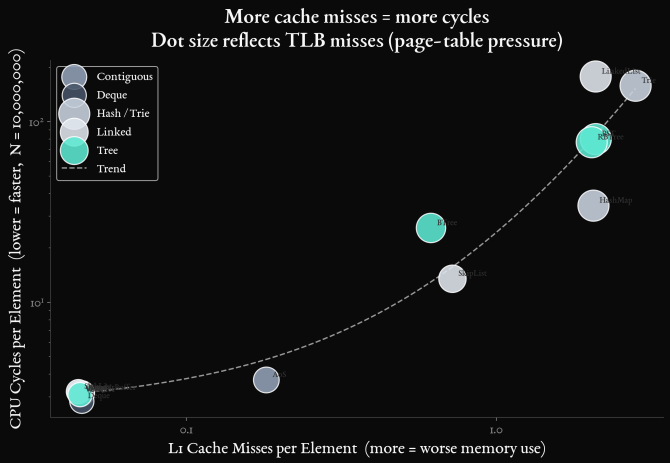
- Modern CPUs operate on a **spatial premise**.
- Contiguous layouts leverage hardware prefetching.
- Pointer-chasing induces serial stalls.
- Memory latency remains the primary bottleneck of 21st-century systems.

Empirical Observations: 16 Primitive Structures



Measured in CPU cycles per element. Sorted by architectural efficiency.

The Causality of Cache Exhaustion



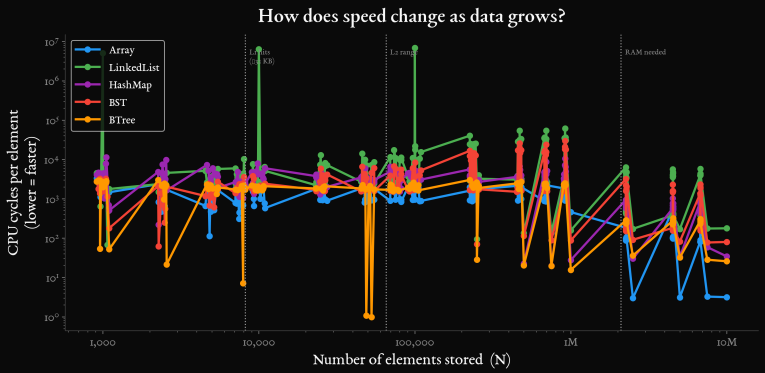
The Linear Drift

The correlation between L1 cache-invalidation and net latency is near-perfect ($R^2 \approx 0.99$).

Observation:

If representation fails the cache, logic cannot recover the speed.

Working-Set Dynamics & Performance Inversion



Log-log scale. Divergence occurs at cache-boundary saturation.

A Proposed Hierarchy of Data Representation

The Ideal
Arrays & Vectors
Optimized for the physical machine.

The Hybrid
BTrees
Clustered nodes to amortize latency.

The Suboptimal
Linked Lists
Architecturally inefficient at scale.

Computational complexity is secondary to spatial complexity on modern hardware.

Synthesis

I. Memory is no longer a passive storage tier, but a performance engine.
II. Representation dictates execution speed more than algorithmics.
III. Contiguity is a prerequisite for high-fidelity systems.

Full dataset available at github.com/SheikhMohsin9311